Z-transform:

$$X(z) = \mathcal{Z}[x_k] = \sum_{k=-\infty}^{\infty} x_k z^{-k}$$

$$X(z) = \mathcal{Z}[x_k] = \sum_{k=0}^{\infty} x_k z^{-k}$$

Z-transform properties:

$$\mathcal{Z}[x((k-j)T)] = \mathcal{Z}[x_{k-j}] = z^{-j}X(z)$$

$$\mathcal{Z}[x((k+j)T)] = \mathcal{Z}[x_{k+j}] = z^{j}X(z) - z^{j}\sum_{i=0}^{j-1} x(iT)z^{-i}$$

$$ax_k + by_k \Leftrightarrow aX(z) + bY(z)$$

Final value theorem:

$$\lim_{k\to\infty} x_k = \lim_{k\to\infty} x(kT) = \lim_{z\to 1}(z-1)X(z)$$

Steady-state value:

$$\lim_{k\to\infty} e(kT) = \lim_{z\to 1}\frac{(z-1)R(z)}{1+G(z)}$$

$$G(z) = \frac{K\prod_{j=1}^{m}(z-z_j)}{(z-1)^N\prod_{k=1}^{p}(z-z_k)}$$

$$e_{ss}[\mathsf{step}] = \frac{1}{1+K_{\mathsf{dc}}}, K_{\mathsf{dc}} = \lim_{z\to 1} G(z)$$

Forward rectangular rule:

$$\frac{U(z)}{E(z)} = \frac{T}{z-1} \Rightarrow s \to \frac{z-1}{T} =: f(z)$$

Backward rectangular rule:

$$\frac{U(z)}{E(z)} = \frac{Tz}{z-1} \Rightarrow s \to \frac{z-1}{Tz} =: f(z)$$

Trapezoidal rule:

$$\frac{U(z)}{E(z)} = \frac{T}{2}\frac{z+1}{z-1} \Rightarrow s \to \frac{2}{T}\frac{z-1}{z+1} =: f(z)$$

Ultimate sensitivity method:

	K_p	T_i	T_d
P	$0.5K_u$		
PI	$0.45K_u$	$P_u/1.2$	
PID	$0.6K_u$	$P_u/2$	$P_u/8$

Backward difference:

$$\frac{dx}{dt} \approx \frac{x(t)-x(t-T)}{T}$$

Forward difference:

$$\frac{dx}{dt} \approx \frac{x(t+T)-x(t)}{T}$$

Central difference:

$$\frac{dx}{dt} \approx \frac{x(t+T)-x(t-T)}{2T}$$

Zero-order hold:

$$G_{\mathsf{ZOH}}(s) = \frac{1-e^{-sT}}{s}$$

Sampling theorem:

$$\omega_s > 2\omega_b \qquad \omega_s = \frac{2\pi}{T}$$

Bandwidth:

If $G(s) = \dfrac{K}{\tau s+1}$, then $\omega_b = \tau^{-1}$

If $G(s) = \dfrac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$,

then $\omega_b = \omega_n\sqrt{1-2\zeta^2+\sqrt{2-4\zeta^2+4\zeta^4}}$

Scheduling:

$$MCT = \mathsf{lcm}(\{p_i\})$$

$$mct = \mathsf{gcd}(\{p_i\})$$

$$2mct - \mathsf{gcd}(mct, p_i) \le d_i$$

Processor utilization:

$$U_p = \sum_{i=1}^{n}\frac{c_i}{p_i}$$

Schedulability:

$$U_p \le 1$$

RM scheduling:

$$\pi_i = \frac{1}{p_i}$$

Basic RMA test:

$$U_p \le U_B(n)$$

$$U_B(n) := n\left(2^{(1/n)}-1\right)$$

Blocking time:

$$B_i = \max_{\forall R_k \text{ used by } \tau_j\in lp(\tau_i)\ s.t.\ PC(R_k)\ge\pi_i}\{\text{execution time of } CS(R_{k,j})\}$$

Extended RMA test:

$$\frac{c_1}{p_1}+\frac{c_2}{p_2}+\cdots+\frac{c_n}{p_n}+\max\left\{\frac{B_1}{p_1},\frac{B_2}{p_2},\cdots,\frac{B_{n-1}}{p_{n-1}}\right\} \le n(2^{1/n}-1)$$

Hyperbolic bound test:

$$\prod_{i=1}^{n}\left(\frac{c_i}{p_i}+1\right) \le 2$$

Response time:

$$w_i^{k+1} = c_i + \sum_{\text{for all } \tau_j\in hp(\tau_i)}\left\lceil\frac{w_i^k}{p_j}\right\rceil c_j$$

Priority ceiling:

$$PC(R) = \max_{\tau_i \text{ that require access to } R}\{\pi_i\}$$

Step:

$$R(z) = \frac{z}{z-1}$$

Ramp:

$$R(z) = \frac{Tz}{(z-1)^2}$$

Quadratic equation:

$$x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$$